VMC Vidyamandir Classes SINCE 1986
IIT JEE | MEDICAL | FOUNDATION

CODE A

JEE Main - 9 | JEE 2022

Date: 24/05/2021 **Maximum Marks: 300**

Timing: 04:00 PM - 09:00 PM **Duration: 3.0 Hrs**

General Instructions

1. The test is of **3 hours** duration.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, ⊖ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the **Invigilator** on duty in the Room/Hall. **However, the candidates are allowed to take away this Test Booklet with them**.
8. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) : ______________________________

Roll Number : ______________________________

OMR Bar Code Number : ______________________________

Candidate's Signature : ____________________ **Invigilator's Signature** ______________

PART I : PHYSICS — 100 MARKS

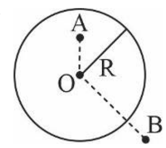
SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. Two spherical conductors A and B having equal radii and carrying equal magnitude of charges in them attract each other with a force F when kept apart at some distance. A third spherical conductor having same radius as that of A but uncharged, is brought in contact with A, then brought in contact with B and finally removed away from both. The new force of attraction between A and B is: {distance between A and B is much more than their radius}

(A) $\frac{F}{4}$ **(B)** $\frac{3F}{4}$ **(C)** $\frac{F}{8}$ **(D)** None of these

2. A thin spherical shell of radius R and charge Q spread uniformly over its surface is as shown in the figure. What is the ratio of potential at point A to the magnitude of Electric field at point B? $\{OA = R/2, OB = 3R\}$



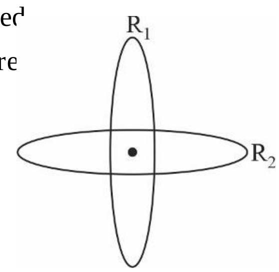
(A) $\frac{7R}{2}$ **(B)** $2R$ **(C)** $9R$ **(D)** $\frac{5}{2}R$

3. The potential of the Electric field at any point (x, y, z) is given by $V = 9x^3 + 3$, where x, y, z are in metres and V is in volts. The intensity of the electric field at (–3, 2, 0) is:

(A) 108 V/m in the positive x direction **(B)** 108 V/m in the negative x direction

(C) 243 V/m in the negative x direction **(D)** 243 V/m in the positive x direction

4. Two concentric rings R_1 and R_2 of same radius R and uniformly charged with charge Q each are arranged perpendicularly as shown in the figure. What is the potential at the centre of the rings?



(A) $\frac{Q}{2\pi\varepsilon_0 R}$ **(B)** $\frac{Q}{2\sqrt{2}\pi\varepsilon_0 R}$

(C) $\frac{\sqrt{2}Q}{\pi\varepsilon_0 R}$ **(D)** $\frac{Q}{4\pi\varepsilon_0 R}$

5. How many excess electrons are there in a body having $-3C$ charge?

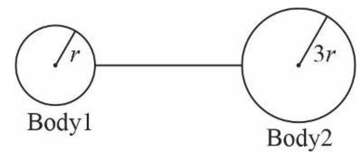
(A) 18.6×10^{18} electrons **(B)** 3 electrons

(C) 10^{18} electrons **(D)** 3.1×10^{18} electrons

6. A point dipole of $\overline{p} = p_0\hat{i}$ is kept at origin. The potential due to this dipole on the y axis at distance d is: (Take $V = 0$ at infinity)

(A) $\frac{-\overline{p}}{4\pi\varepsilon_0 d^2}$ **(B)** $\frac{|\overline{p}|}{4\pi\varepsilon_0 d^2}$ **(C)** Zero **(D)** $\frac{2|\overline{p}|}{4\pi\varepsilon_0 d^2}$

7. Two spherical conductors of radii r and $3r$ are kept at a large distance from each other. They contain charges $2q$ and $-q$ respectively. If they are connected with a wire, which of the following observations is true?



(A) Charge $\frac{3q}{4}$ flows from Body 1 to Body 2

(B) Charge $\frac{-7q}{4}$ flows from Body 2 to Body 1

(C) Charge $\frac{3q}{4}$ flows from Body 2 to Body 1

(D) No charge flows through the wire

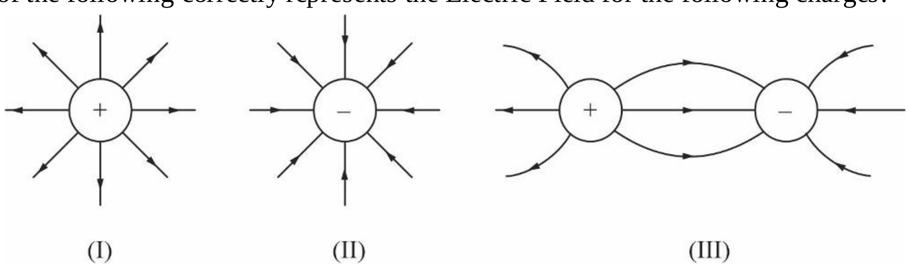
8. An electric field of 100N/C is applied to an electric dipole at an angle of 60°. The value of electric dipole moment is $10^{-10}\ Cm$. What is the potential energy of the electric dipole?

(A) $5\times10^{-9}\,J$ **(B)** $-5\times10^{-9}\,J$ **(C)** $5\sqrt{3}\times10^{-9}\,J$ **(D)** $-5\sqrt{3}\times10^{-9}\,J$

9. Which of the following statement is true?

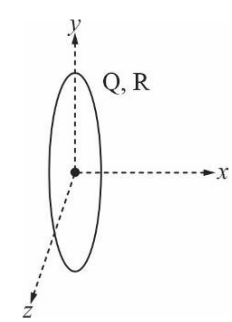
(A) Electric potential decreases in the direction of electric field
(B) Electric field lines end at positive charge
(C) Force on charge is always in the direction of electric field
(D) Electric field lines intersect near the charges

10. Which of the following correctly represents the Electric Field for the following charges?



(A) Only I and II **(B)** Only I **(C)** Only II **(D)** All I, II and III

11. A thin uniformly charged ring of radius R is kept fixed in the y-z plane with centre at origin as shown in figure. If a charge $-q$ of mass m is left from coordinate $(x, 0, 0)$, which of the following statements is correct? $\{R >>>> x\}$



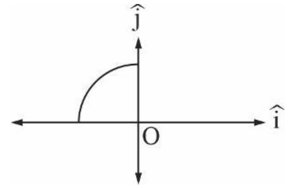
(A) Charge –q will cross the origin and reach ∞ on negative x side

(B) Charge –q will do SHM with time period proportional to $\sqrt{Qq}$

(C) Charge –q will do SHM with time period proportional to $\frac{1}{\sqrt{Qq}}$

(D) Charge –q will never reach coordinate (0, 0, 0)

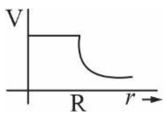
12. A thin quarter ring of radius 2R has a positive charge $\frac{Q}{2}$ distributed uniformly over it. The net field E at centre O is:

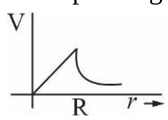


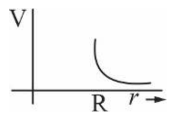
(A) $\frac{Q}{4\pi^2\varepsilon_0 R}\hat{i} - \frac{Q}{4\pi^2\varepsilon_0 R}\hat{j}$ **(B)** $\frac{2Q}{\pi^2\varepsilon_0 R}\hat{i} - \frac{2Q}{\pi^2\varepsilon_0 R}\hat{j}$

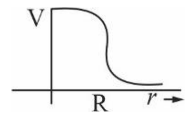
(C) $\frac{Q}{8\pi^2\varepsilon_0 R}\hat{i} - \frac{Q}{8\pi^2\varepsilon_0 R}\hat{j}$ **(D)** $\frac{Q}{8\pi\varepsilon_0 R}\hat{i} - \frac{Q}{8\pi\varepsilon_0 R}\hat{j}$

13. In an uniformly non conducting sphere of total charge Q and radius R, potential V is plotted as function of distance from the centre. The corresponding graph will be:

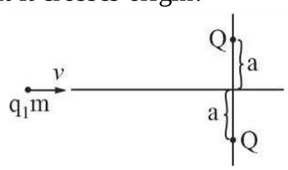
(A) 

(B) 

(C) 

(D) 

14. A charged particle q of mass m is shot towards origin with a speed v from a very large distance as shown in the figure. Two charges Q each are fixed at $y = a$ and $y = -a$ respectively. Find the minimum velocity v which should be given to q such that it crosses origin?



(A) $\left(\frac{Qq}{\pi\varepsilon_0 ma}\right)^{1/2}$ **(B)** $\left(\frac{4Qq}{\pi\varepsilon_0 ma}\right)^{1/2}$ **(C)** $\left(\frac{2Qq}{\pi\varepsilon_0 ma}\right)^{1/2}$ **(D)** None of the above

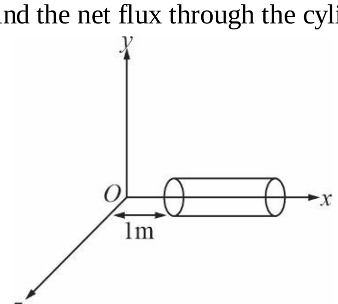
15. A thin spherical conducting shell of radius R has a charge q. Another charge $q/2$ is placed at the centre of the shell. The electrostatic potential at a distance $R/3$ from the centre of the shell is:

(A) $\frac{5q}{8\pi\varepsilon_0 R}$ **(B)** $\frac{q}{4\pi\varepsilon_0 R}$ **(C)** $\frac{3q}{4\pi\varepsilon_0 R}$ **(D)** $\frac{2q}{\pi\varepsilon_0 R}$

16. An electric dipole is formed by two equal and opposite *charges* of magnitude q with separation d. They are kept in a uniform Electric field. What is the magnitude of force on the dipole if dipole moment makes an angle of 30° with Electric field?

(A) $\frac{qdE}{2}$ **(B)** qdE **(C)** $2qE$ **(D)** Zero

17. A cylindrical box of length 1m and area of cross-section $25cm^2$ is placed in a three dimensional coordinate system as shown in the figure. The electric field in the region is given by $\vec{E} = 50x\hat{i}$ where E is in NC^{-1} and x is in metres. Find the net flux through the cylinder.

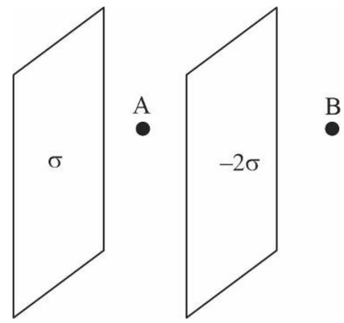


(A) $0.125 Nm^2/C$ **(B)** $0.375 Nm^2/C$ **(C)** $0.25 Nm^2/C$ **(D)** Zero

18. Assume that an Electric Field $E = 20y^3\hat{j}$ exists in space. Then, the potential difference $V_A - V_0$, where V_0 is the potential at the origin and V_A is the potential at $y = 1m$ is:

(A) 5V **(B)** 20V **(C)** $-20V$ **(D)** $-5V$

19. Two fixed large insulating plates uniformly charged are kept at a small distance as shown in the figure. What is the ratio of electric field intensity at point A to that of point B?



(A) $\frac{1}{3}$

(B) 3

(C) 6

(D) $\frac{2}{3}$

20. Two charges apply a force F on each other when they are kept in vacuum at a distance of d. If they are kept in a medium of dielectric 3 at a distance of 2d, how much force will they feel?

(A) $\frac{F}{12}$ **(B)** $\frac{3F}{4}$ **(C)** $\frac{4F}{3}$ **(D)** $12F$

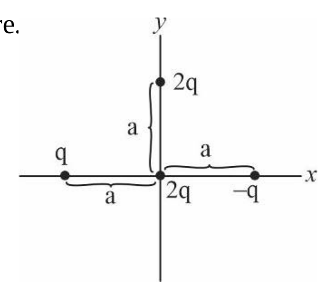
SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

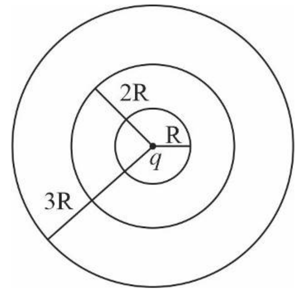
21. Four charges are placed at x and y axes as shown in the figure. The force on the charge kept at origin is $\frac{Kq^2}{a^2}\sqrt{n}$. Find n.

$\left\{K = \frac{1}{4\pi\varepsilon_0}\right\}$

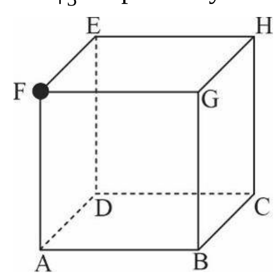


22. Let $\rho(r) = \rho_0 r^2$ be the charge distribution for a solid sphere of radius R, where r is the distance from the centre. The magnitude of Electric field at a point which is at the distance of 3R from the centre of the sphere is $\frac{\rho_0 R^3}{n\varepsilon_0}$. Find n.

23. A charge q is kept at the centre of three thin concentric spherical conducting shells of radius $R, 2R$ and $3R$ respectively as shown in the figure. Find the ratio of charge density of Outer surface of innermost shell to Outer surface of outermost shell.



24. A charge q is kept at the vertex F of a cube as shown in the figure. Let flux passing through surface ABGF, BCHG and HCDE be ϕ_1, ϕ_2 and ϕ_3 respectively.



Find $\dfrac{\phi_2 - \phi_1}{\phi_3 - \phi_1}$.

25. On moving a charge of 15C by $3m$, 30J of work is done, then the potential difference between the points is nV. Find $|4n|$. {| | is the modulus function}

SPACE FOR ROUGH WORK

PART II : CHEMISTRY — 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. In a solid AB, having the NaCl structure, 'A' atoms occupy the corners of cubic unit cell. If all the face centred atoms along one of the axis are removed, then the resultant stoichiometry of solid is

(A) AB_2 **(B)** A_2B **(C)** A_4B_3 **(D)** A_3B_4

2. A certain metal P has FCC structure with unit cell length $3.16\,\text{Å}$, the size of largest atom that can be accommodated into the interstices of the metal without changing the position of metal atoms is

(A) 0.53Å **(B)** 0.88Å **(C)** 0.149Å **(D)** None of the above

3. In diamond, the lattice is FCC with C atoms occupying lattice points as well as some of the voids. Which of the following statement(s) is/are correct about the diamond structure?

(I) In the diamond lattice, half of the tetrahedral voids are occupied.

(II) Percentage space occupied (packing efficiency) is 74%.

(III) In the diamond lattice all octahedral voids are occupied.

(A) Only I **(B)** Only III **(C)** I, II and III **(D)** Only II

4. The unit cell length of NaCl is observed to be 0.5627 nm by X-ray diffraction studies. The measured density of NaCl is 2.164 g/cm^3. What is the percentage of missing Na^+ & Cl^- ions?

(A) 0.775 **(B)** 0.225 **(C)** 0.683 **(D)** 0.414

5. Frenkel defect is not found in the halides of alkali metals because alkali metals have

(A) high electro positivity **(B)** high ionic radii

(C) high reactivity **(D)** ability to occupy interstitial sites

6. A compound ABC forms cubic close packed structure. In the lattice, A occupies the lattice points and B and C occupy the alternate tetrahedral holes. If all the ions along one body diagonal are removed, then formula of the compound will be:

(A) $A_{3.5}B_3C_3$ **(B)** $A_{3.75}B_3C_3$ **(C)** $A_2B_3C_3$ **(D)** $A_{3.25}B_3C_3$

7. In which type of voids magnesium and silicon is present respectively in asbestos [$Mg_3Si_2O_5$] if $r_{Mg^{+2}}/r_{O^{2-}} = 0.59$ and $r_{Si^{+4}}/r_{O^{2-}}$ is 0.29.

(A) tetrahedral & octahedral **(B)** both tetrahedral

(C) octahedral & tetrahedral **(D)** both octahedral

8. The yellow colour and conducting nature produced on heating of ZnO is due to

(A) metal excess defect

(B) extra positive ions present in an interstitial site

(C) trapped electrons

(D) All of the above

9. In a compound AB, the ionic radii of A^+ and B^- are 88 pm and 200 pm respectively. The coordination number of A^+ is

(A) 6 **(B)** 8 **(C)** 4 **(D)** 12

10. Which one of the following is true for Schottky defect in crystals?

(A) Schottky defects increases the density of the crystal.

(B) Schottky defect is found in crystal having large $\frac{r^+}{r^-}$ ratio.

(C) Schottky defects are non thermodynamic defects.

(D) Schottky defect is shown by highly ionic compounds having high coordination number.

11. An ionic solid is HCP of Q^{2-} ions and P^{x+} ions are in half of the tetrahedral voids. The value of x should be:

(A) 1 **(B)** 2 **(C)** 4 **(D)** 1/2

12. The composition of a sample of cuprous oxide is found to be $Cu_{1.92}O_{1.00,}$ due to metal deficient defect. The molar ratio of Cu^{2+} and Cu^+ ions in the crystal is:

(A) 4 : 5 **(B)** 1 : 12.5 **(C)** 1 : 23 **(D)** 1 : 24

13. what is the void space per unit cell for metallic silver crystallizing in the FCC system, the edge length of the unit cell being 4Å ?

(A) 47.36Å^3 **(B)** 30.72Å^3 **(C)** 20.48Å^3 **(D)** 16.64Å^3

14. A solid element (monoatomic) exists in cubic crystal. If its atomic radius is 1.0Å and the ratio of packing fraction and density is $0.1 cm^3/g$, then the atomic mass of the element is $(N_A = 6\times10^{23})$

(A) 8π **(B)** 16π **(C)** 80π **(D)** 4π

15. A semiconductor $Y_2Ba_2Cu_3O_7$ is prepared by a reaction involving Y_2O_3, BaO_2 and CuO. The ratio of their moles required should be:

(A) 1 : 2 : 4 **(B)** 1 : 2 : 3 **(C)** 3 : 2 : 1 **(D)** 1 : 1.5:2.5

16. Metallic gold crystallizes in FCC lattice with edge-length 4.07Å. The closest distance between gold atoms is:

(A) 3.525Å **(B)** 5.714Å **(C)** 2.877Å **(D)** 1.428Å

17. A match box exhibits___________ geometry.

(A) cubic **(B)** orthorhombic **(C)** triclinic **(D)** monoclinic

18. In a tetragonal crystal

(A) $\alpha=\beta=90°\neq\gamma; a=b=c$ **(B)** $\alpha=\beta=\gamma=90°; a=b\neq c$

(C) $\alpha=\beta=\gamma=90°; a\neq b\neq c$ **(D)** $\alpha=\beta=90°; \gamma=120°; a=b\neq c$

19. Sodium oxide has anti-fluorite structure. The percentage of the tetrahedral voids occupied by the sodium ions is:

(A) 12% **(B)** 25% **(C)** 50% **(D)** 100%

20. The number of tetrahedral and octahedral voids in hexagonal close packing unit cell is:

(A) 8, 4 **(B)** 2, 1 **(C)** 12, 6 **(D)** 6, 12

SECTION-2

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21. The number of unit cells that are present in 78 gm of potassium that crystallises in Body centred cubic crystal is $x \times 10^{23}$. What is the value of x, (Given: Atomic mass of K = 39, $N_A = 6 \times 10^{23}$)

22. The density of crystalline sodium chloride is 2.167 g/cm^3. What would be the side length (in cm) of cube containing one mole of NaCl?

23. An element (atomic mass = 125) crystallizes in simple cubic structure. If the diameter of the largest sphere which can be placed in the crystal without disturbing the crystal is 366 pm & the density of crystal is 'd', then the value of 3d is?

24. A solid AB has NaCl type structure. If radius of the cation A^+ is 120 pm. Then the value of radius (in pm) of B^- ion is?

25. The number of atoms in 100 gm of an FCC crystal with density (d) = 10 g/cm^3 & cell edge as 200 pm is $x \times 10^y$. Then what is the value of (xy/10)? [x is a single integer digit]

SPACE FOR ROUGH WORK

PART III : MATHEMATICS 100 MARKS

SECTION-1

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

1. The domain of the function $f(x)=\log_{\left[x+\frac{1}{2}\right]}(2x^2+x-1)$ where $[\cdot]$ denotes greatest integer function :

(A) $\left[\frac{3}{2},\infty\right)$ **(B)** $(2,\infty)$ **(C)** $\left(-\frac{1}{2},\infty\right)-\left\{\frac{1}{2}\right\}$ **(D)** $\left(\frac{1}{2},1\right)\cup(1,\infty)$

2. Let $f:R-\left\{\frac{3}{2}\right\}\to R,\ f(x)=\frac{3x+5}{2x-3}$. Let $f_1(x)=f(x)$ and $f_n(x)=f(f_{n-1}(x))$ for $n\ge 2, n\in N$, then $f_{2008}(x)+f_{2009}(x)$ is :

(A) $\frac{2x^2+5}{2x-3}$ **(B)** $\frac{x^2+5}{2x-3}$ **(C)** $\frac{2x^2-5}{2x-3}$ **(D)** $\frac{x^2-5}{2x-3}$

3. If $f(x)=\sin\left(\log\left(\frac{\sqrt{4-x^2}}{1-x}\right)\right); x\in R,$ then range of $f(x)$ is given by :

(A) $[0,1]$ **(B)** $(-1,1)$ **(C)** $[-1,1]$ **(D)** $(-2, 1)$

4. The number of integral values of x in the domain of function f defined as

$f(x)=\sqrt{\ln|\ln|x||}+\sqrt{7|x|-|x|^2-10}$ is :

(A) 5 **(B)** 6 **(C)** 7 **(D)** 8

5. Let $f(x)=\frac{x}{\sqrt{1+x^2}}$, then $\underbrace{fofof\ ....\ of}_{n\text{ times}}(x)$ is :

(A) $\frac{x}{\sqrt{1+\left(\sum_{r=1}^{n} r\right)x^2}}$ **(B)** $\frac{x}{\sqrt{1+\left(\sum_{r=1}^{n} 1\right)x^2}}$ **(C)** $\left(\frac{x}{\sqrt{1+x^2}}\right)^n$ **(D)** $\frac{nx}{\sqrt{1+nx^2}}$

6. Which of the following is periodic with fundamental period π ?

(A) $f(x)=\cos x+\left[\left|\frac{\sin x}{2}\right|\right]$ where [.] denotes G.I.F.

(B) $g(x)=\frac{\sin x+\sin 7x}{\cos x+\cos 7x}+|\sin x|$

(C) $h(x)=\{x\}+|\cos x|$ where {.} denotes fractional part function

(D) $\phi(x)=|\cos x|+\ln(\sin x)$

7. Let $a=\lim_{x\to 0}\frac{\ln(\cos 2x)}{3x^2}, b=\lim_{x\to 0}\frac{\sin^2 2x}{x(1-e^x)}, c=\lim_{x\to 1}\frac{\sqrt{x}-x}{\ln x}$ then :

(A) $a<b<c$ **(B)** $b<c<a$ **(C)** $a<c<b$ **(D)** $b<a<c$

8. Let $f(x)=\begin{cases}|x-2|+a^2-6a+9 & , x<2\\ 5-2x & , x\geq 2\end{cases}$

If $\lim\limits_{x\to 2}[f(x)]$ exists, the value a cannot take is (where [.] represents the G.I.F.)

(A) 2 **(B)** $\frac{5}{2}$ **(C)** 3 **(D)** $\frac{7}{2}$

9. Let $f:R\to R$ $f(x)=\begin{cases}(-1)^n, \text{ if } x=\frac{1}{2^{2^n}}, n=1,2,3,\ldots\\ 0, \quad \text{otherwise}\end{cases}$ then identify the correct statement

(A) $\lim\limits_{x\to 0^+} f(x)=1$ **(B)** $\lim\limits_{x\to 0} f(x)$ exists

(C) $\lim\limits_{x\to 0} f(x)\cdot f(2x)=0$ **(D)** $\lim\limits_{x\to 0} f(x)\cdot f(2x)$ does not exist

10. Let $f(x)=\begin{cases}(1+ax)^{1/x} & , \quad x<0\\ b & , \quad x=0\\ \dfrac{(x+c)^{1/3}-1}{(x+1)^{1/2}-1} & , \quad x>0\end{cases}$

And $f(0^-)=f(0)=f(0^+)$ then $3(e^a+b+c)$ is equal to :

(A) 3 **(B)** 6 **(C)** 7 **(D)** 8

11. If $f(x)=\begin{cases}\dfrac{ae^{\sin x}+be^{-\sin x}-c}{x^2} & , x\neq 0\\ 2 & , x=0\end{cases}$ and $\lim\limits_{x\to 0} f(x)=f(0)$ then

(A) $a=b=c$ **(B)** $a=2b=3c$ **(C)** $a=b=2c$ **(D)** $2a=2b=c$

12. Which of the following is indeterminate from : (where [.] denotes G.I.F.)

(A) $\lim\limits_{x\to 0}\sqrt{x}\sin\left(\dfrac{e^{x^2}}{x^3}\right)$ **(B)** $\lim\limits_{x\to 2^+}\dfrac{[x]-2}{x^2-4}$

(C) $\lim\limits_{x\to \pi/2}\left(x-\dfrac{\pi}{2}\right)\cos x$ **(D)** $\lim\limits_{x\to 0^+}\dfrac{\sin(x-[x])}{x-[x]}$

13. Let $f(x)=\begin{cases}\dfrac{3x-x^2}{2}, & x<2\\ [x-1]\ , & 2\leq x<3\\ x^2-8x+17\ , & x\geq 3\end{cases}$

Then which of the following does not hold good?

(A) $\lim\limits_{x\to 2} f(x)=1$ **(B)** $\lim\limits_{x\to 3^-} f(x)=2$ **(C)** $\lim\limits_{x\to 3^+} f(x)=2$ **(D)** None of these

14. If $f(x)=\begin{cases}x^3 & , x\in Q\\ -x^3 & , x\notin Q\end{cases}$ then incorrect statement is :

(A) $f(x)$ is non periodic **(B)** $f(x)$ is many-one

(C) $f(x)$ is one one **(D)** range of $f(x)$ is R

15. Let $f(x)=\begin{cases} x^2 & , \quad 0<x<2 \\ 2x-3 & , \quad 2\le x<3 \\ x+2 & , \quad x\ge 3 \end{cases}$ then which among the followings is not true.

(A) $f\left(f\left(f\left(\frac{3}{2}\right)\right)\right)=f\left(\frac{3}{2}\right)$ **(B)** $1+f\left(f\left(f\left(\frac{5}{2}\right)\right)\right)=f\left(\frac{5}{2}\right)$

(C) $f\left(f\left(f(2)\right)\right)=f(1)$ **(D)** $f(f(f(f(3))))=9$

16. $\lim\limits_{x\to\pi/3}\dfrac{\sin\left(\frac{\pi}{3}-x\right)}{2\cos x-1}$ is equal to :

(A) $\dfrac{2}{\sqrt{3}}$ **(B)** $\dfrac{1}{\sqrt{3}}$ **(C)** $\sqrt{3}$ **(D)** $\dfrac{1}{2}$

17. $\lim\limits_{x\to 0}\left(1+\dfrac{a\sin bx}{\cos x}\right)^{1/x}$ where a, b are non-zero constants is equal to :

(A) $e^{a/b}$ **(B)** ab **(C)** e^{ab} **(D)** $e^{b/a}$

18. For $n\in N$, let $f_n(x)=\tan\dfrac{x}{2}(1+\sec x)(1+\sec 2x)(1+\sec 4x)....(1+\sec 2^n x)$ then $\lim\limits_{x\to 0}\dfrac{f_n(x)}{2x}$ is equal to :

(A) 0 **(B)** 2^n **(C)** 2^{n-1} **(D)** 2^{n+1}

19. $\lim\limits_{x\to 0}\dfrac{2e^{2\sin x}-e^{\sin x}-1}{(x^2+2x)}$ equals to :

(A) $\dfrac{3}{2}$ **(B)** $e^{3/2}$ **(C)** 2 **(D)** e^2

20. If [.] denotes G.I.F. and $f(x)=\dfrac{5\sin[\cos x]}{[\cos x]+2}$ then $\lim\limits_{x\to\frac{\pi}{2}^-}f(x)+\lim\limits_{x\to\frac{\pi}{2}^+}f(x)$ is :

(A) 0 **(B)** $5\sin(1)$ **(C)** $-5\sin(1)$ **(D)** $\dfrac{5}{2}\sin(1)$

SPACE FOR ROUGH WORK

SECTION-2

This Section contains 5 Numerical Value Type Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the ⊕ sign for positive values. However, for negative values, Θ sign should be bubbled*. (Example: 6, 81, 1.50, 3.25, 0.08)

21. Consider the function $f(x)=\begin{cases}\dfrac{\max\left(x,\frac{1}{x}\right)}{\min\left(x,\frac{1}{x}\right)} & ,\text{ if } x\neq 0\\ 1 & ,\text{ if } x=0\end{cases}$ then $\lim\limits_{x\to 0^-} f(x)+\lim\limits_{x\to 1^-} f(x)+\lim\limits_{x\to -1^+} f(x)$ is :

22. If $\lambda=\lim\limits_{x\to 3^-}\dfrac{(e^{(x+3)\ln 27})^{x/27}-9}{3^x-27}$ then value of $\dfrac{1}{\lambda}$ is :

23. If $f(x)$ is an even periodic function with period 10. In [0, 5], $f(x)=\begin{cases}2x & , \quad 0\le x<2\\ 3x^2-8 & , \quad 2\le x<4\\ 10x & , \quad 4\le x\le 5\end{cases}$

then $\dfrac{1}{10}[f(-4)+f(-13)+f(-11)-1]$ is :

24. Let $u_n=\sum\limits_{r=1}^{n}\dfrac{r}{2^r}, n\ge 1$ then $\lim\limits_{n\to\infty} u_n$ is

25. The value of $\lim\limits_{x\to\frac{\pi}{4}}(1+[x])^{\frac{1}{\ln(\tan x)}}$ is : (where [.] denotes G.I.F.)

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End of JEE Main - 9 | JEE 2022